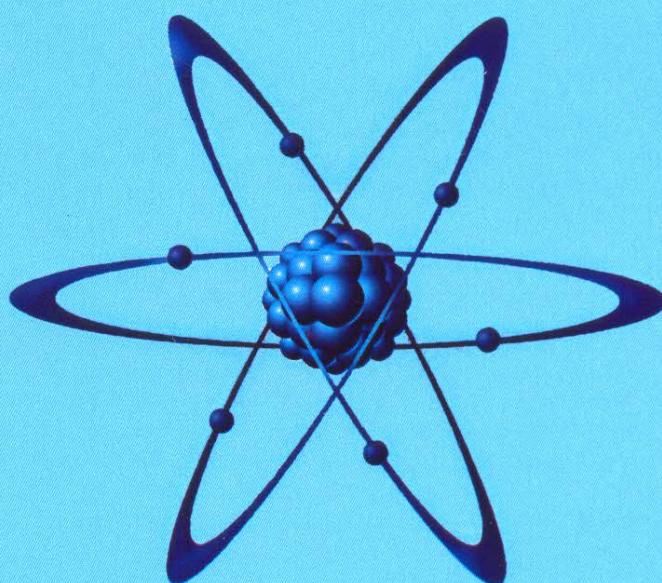


International Scientific and Practical
Conference

WORLD Science



TOPICAL RESEARCHES
OF THE WORLD SCIENCE

Vol. IV



June 20-21, 2015
DUBAI, UAE

ROST PUBLISHING



**EXPO 2020
DUBAI, UAE**

CAND DATE CITY

But Supply ...



INTERNATIONAL SCIENTIFIC AND PRACTICAL CONFERENCE

WORLD Science

Proceedings of the conference

**TOPICAL RESEARCHES OF THE
WORLD SCIENCE**
(June 20-21, 2015)

Vol.IV

Rost Publishing
Dubai
2015

- Printing.
- Renewable energy sources (RES).

In the future, it is planned to introduce the operational leasing, which will have a significant impact on the economy as a whole. However, the leasing market still does not have this type of lease.

Also be aware that the expected rapid growth of Uzbekistan's economy and growth of the leasing sector will not have much impact on the share of leasing in the GDP of the Republic. This does not mean that leasing sector will stop its development, but the fact that Uzbekistan's economy is sufficiently developed, whose growth is diversified and the impact of leasing it significantly reduced.

Today, leasing is a fast growing segment of the economy of the Republic, and there are many favoring factors that contribute to its rapid growth in the near future. Leasing market participants are optimistic about the prospects of the industry, its transparency, and competitiveness. In the future, this service will be a key element in the implementation of the innovation economy and can take a strong position among financial sources allocated for the modernization and creation of new competitive industries of the Republic of Uzbekistan.

REFERENCES

1. Zafar B. Mustafaev: "Overview of leasing industry in Uzbekistan in the 1st half 2014", Uzbekistan Lessors Association, Tashkent 2014, S. 8.

Dr. Eng. (Doctor of Engineering) Pil E. A.

St. Petersburg State Marine Technical University, St. Petersburg, Russia

THEORY OF THE FINANCIAL CRISES

There is the theory of financial crises, which are constantly evolving in modern society and cover more or less almost all countries of the world, proposed in this article. Under the calculations made, the author has built two and three-dimensional graphics that show more accurately the influence of various parameters on the country's GDP (gross domestic product). The performed calculations made it possible to build a table and identify scenarios for overcoming the financial crisis.

Earlier in his article the author has shown that the shell theory can be applied for describing the economic processes, which are taking place in the

economy of the country [2]. And besides the author has also described in his articles, the borders of the existence of financial shells of small, medium-sized and big business, which can be affected by both external and internal forces [1, 3].

Gross domestic product can be calculated by the three following methods:

1. as the sum of gross value added (production method);
2. as the sum of end-use components (end-use method);
3. as the sum of primary income (distribution method).

Further a variety of parameters such as consuming capacity, investment, government spending, exports, imports and others can be used in the calculations. The formula for calculating the volume of financial shell V_{euc} , which corresponds to the country's GDP_c i.e. $V_{euc}(GDP_c) = f(X1, X2, X3, X4)$ was used within making calculations in the following article. Here $X1, X2, X3$ and $X4$ are the parameters affecting the country's GDP_c . It should immediately be noted that during calculation and plotting of construction drawings, the parameters of $X1, X2, X3$ and $X4$ could be constant values, increase or decrease by 10 times. On the basis of the calculations made, 72 graphics were built, which can be divided into the four following groups:

- parameter values $X1, X2, X3$ and $X4$ increase and are constant;
- parameter values $X1, X2, X3$ and $X4$ decrease and are constant;
- parameter values $X1, X2, X3$ and $X4$ decrease and increase;
- parameter values $X1, X2, X3$ and $X4$ are constant, they decrease and increase.

The Fig. 1 shows two-dimensional $V_{euc}(GDP_c)$ dependency graph when $X1 = X3 = 1, X2 = 0.1 \dots 0.99, X4 = 1 \dots 10$, which shows that the V_{euc} values first decrease by 11.4 times, and then increase by 6.3 times. Fig. 2 shows two 3D-graphics that give a possibility to visualize the changes of V_{euc} . In this case it is appropriate to have the values of extreme points, as at them the value of V_{euc} i.e. GDP_c is the highest.

The following figure 3, shows that when $X1 = 1, X2 = 0.99 \dots 0.1, X3 = X4 = 1 \dots 10$, the plotted curve V_{euc} decreases at the point 2 and then it increases gradually by 3.4 times. Fig. 4 shows two views of this dependence in the form of three-dimensional graphs. In this example, we have to take the values of the parameters at the point 10, at which there is the maximum volume of the financial shell.

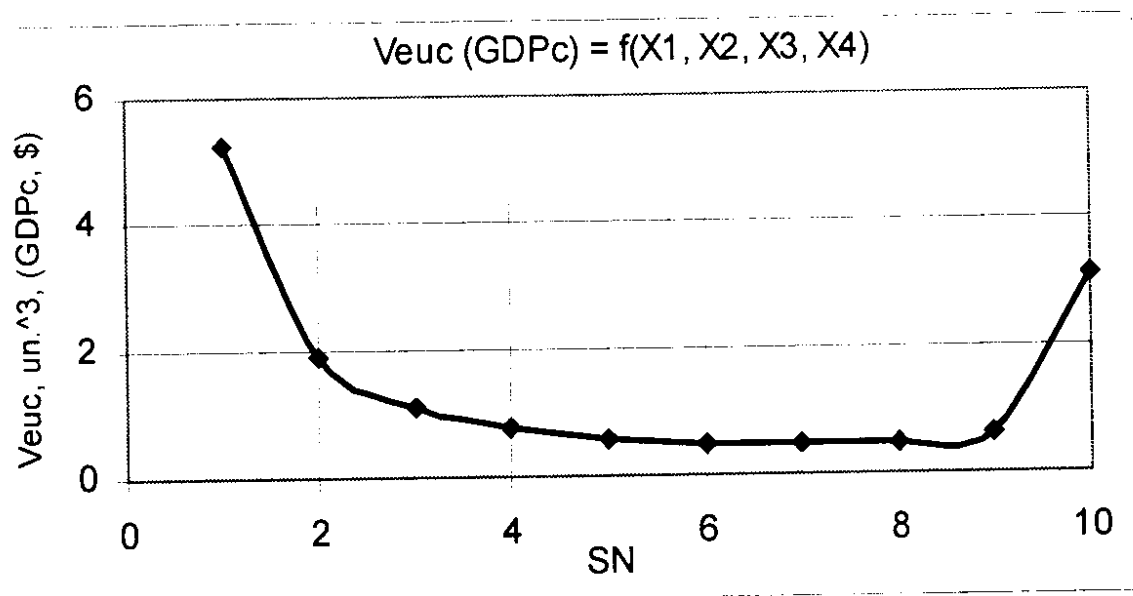
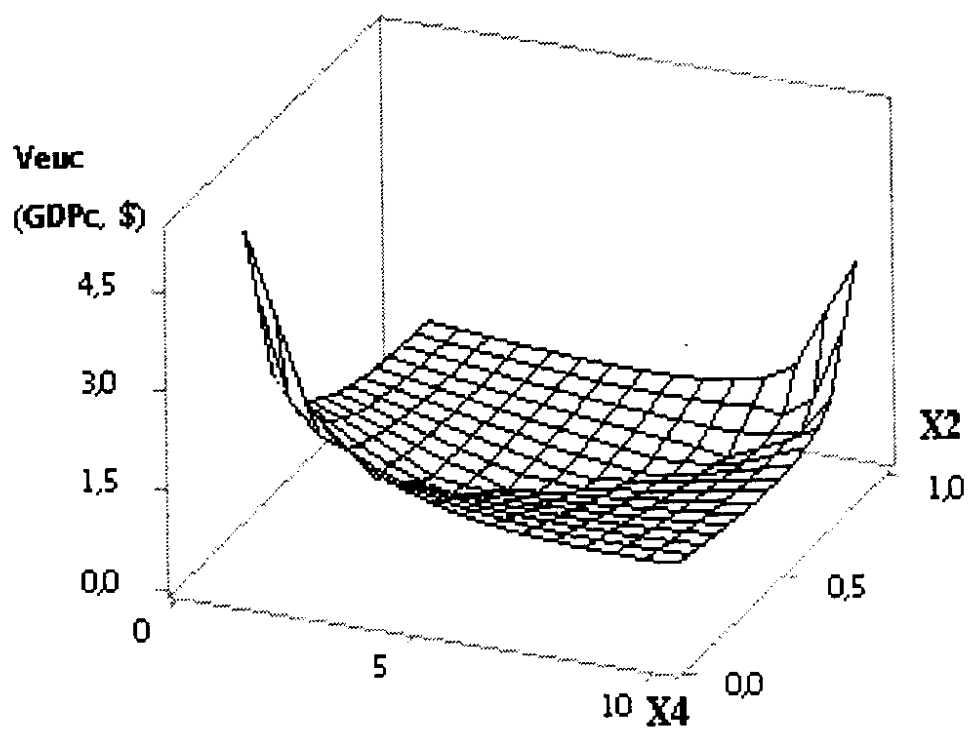
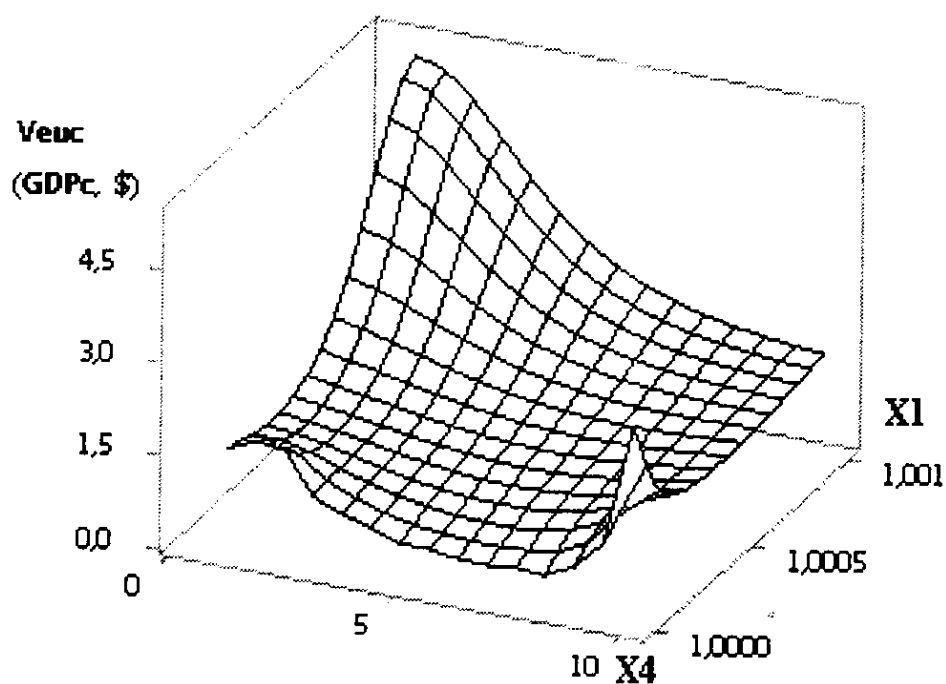


Fig. 1 Dependence $Veuc (GDPc) = f(X1, X2, X3, X4)$

when $X1 = X3 = 1$, $X2 = 0,1...0,99$, $X4 = 1...10$



a



b

Fig. 2 3D-graphics:

a – $V_{euc} (GDP_c) = f(X4, X2)$; *b* – $V_{euc} (GDP_c) = f(X4, X1)$

when $X1 = X3 = 1$, $X2 = 0,1...0,99$, $X4 = 1...10$

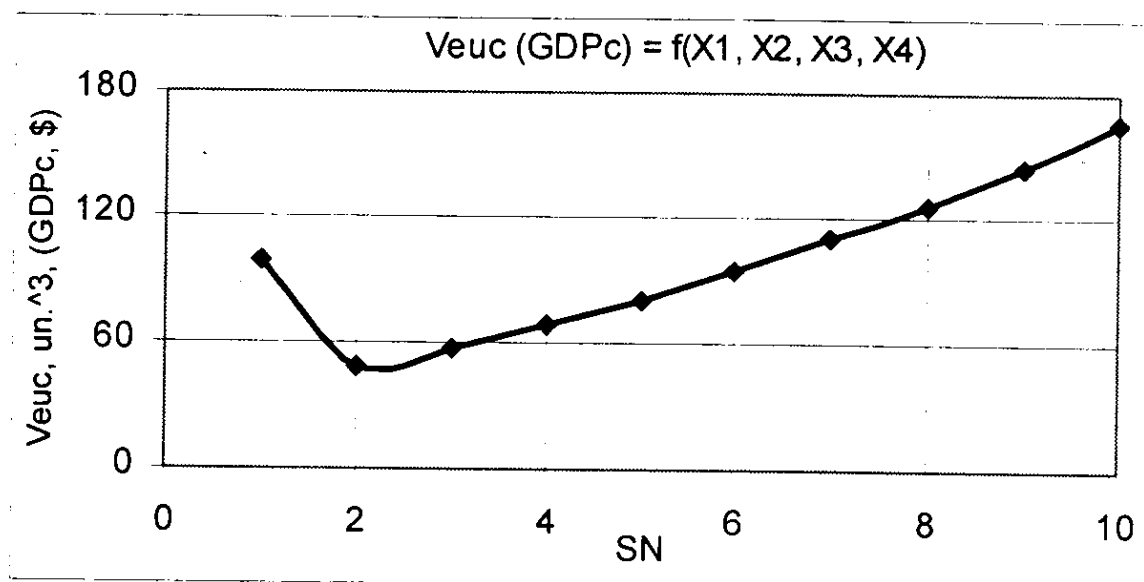
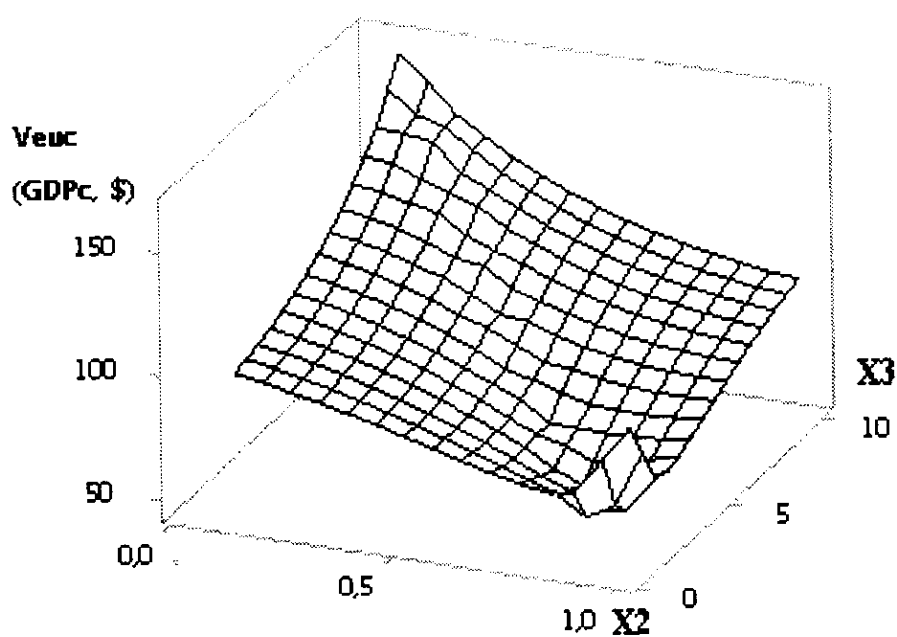
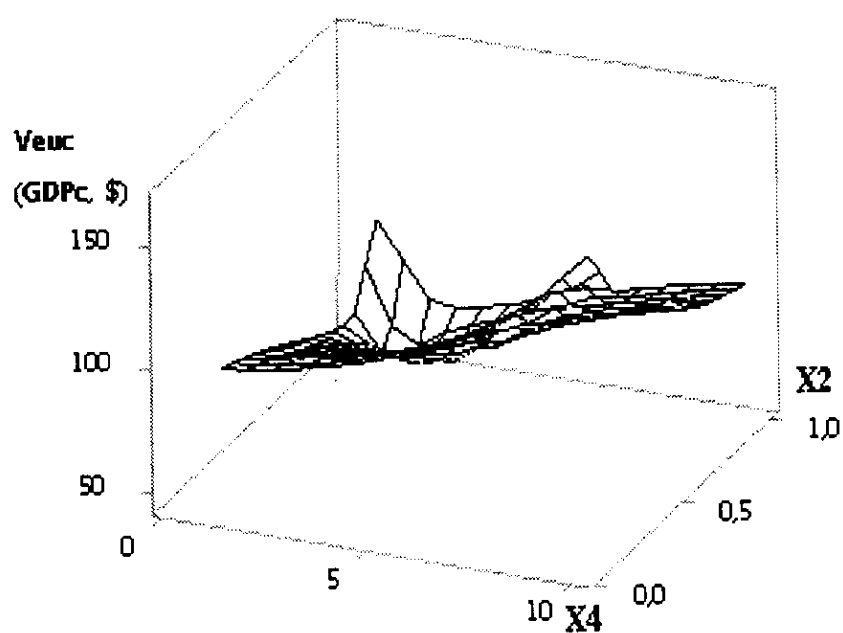


Fig. 2 Dependence $V_{euc} (GDP_c) = f(X1, X2, X3, X4)$

when $X1 = 1$, $X2 = 0,1...0,99$, $X3 = X4 = 1...10$



a



b

Fig. 4 3D-graphics:

$$a - V_{euc} (GDP_c) = f(X2, X3); b - V_{euc} (GDP_c) = f(X4, X2)$$

when $X1 = 1$, $X2 = 0,1...0,99$, $X3 = X4 = 1...10$

In the Fig. 5 we can see that here the plotted V_{euc} curve has a maximum at the point 4 after which its values decrease. Hence we can conclude that it's advantageous for us, to have the following parameter values: $X1 = 4$, $X2 = 0.4$, $X3 = 0.7$, and $X4 = 1$, because in this case, the economy of the analyzed country will have the maximum value of V_{euc} . Three-dimensional image of the curve V_{euc} , plotted in the Fig. 5 is shown in the form of four 3D surfaces in the Fig. 6.

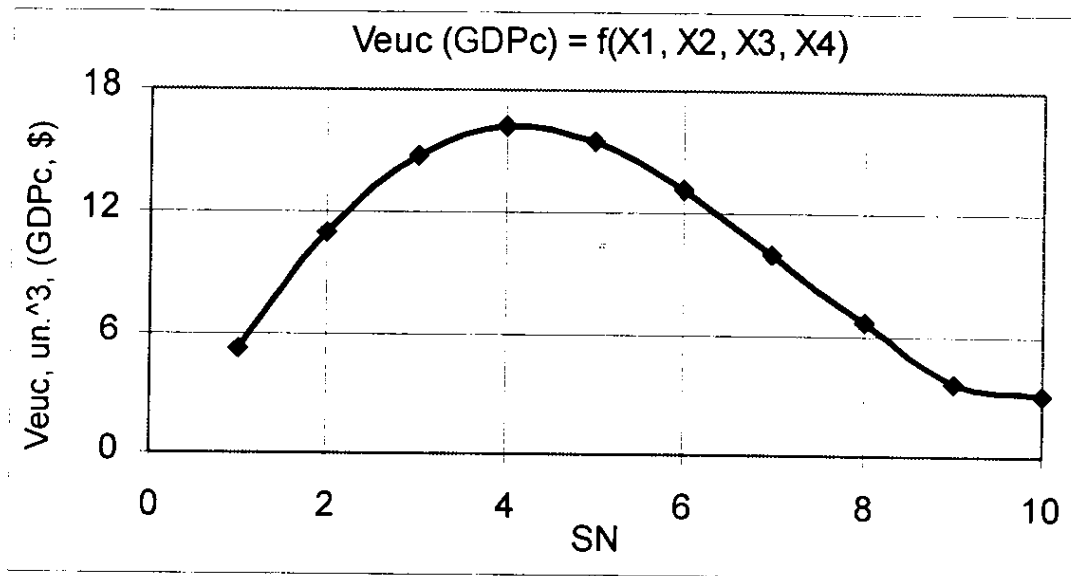
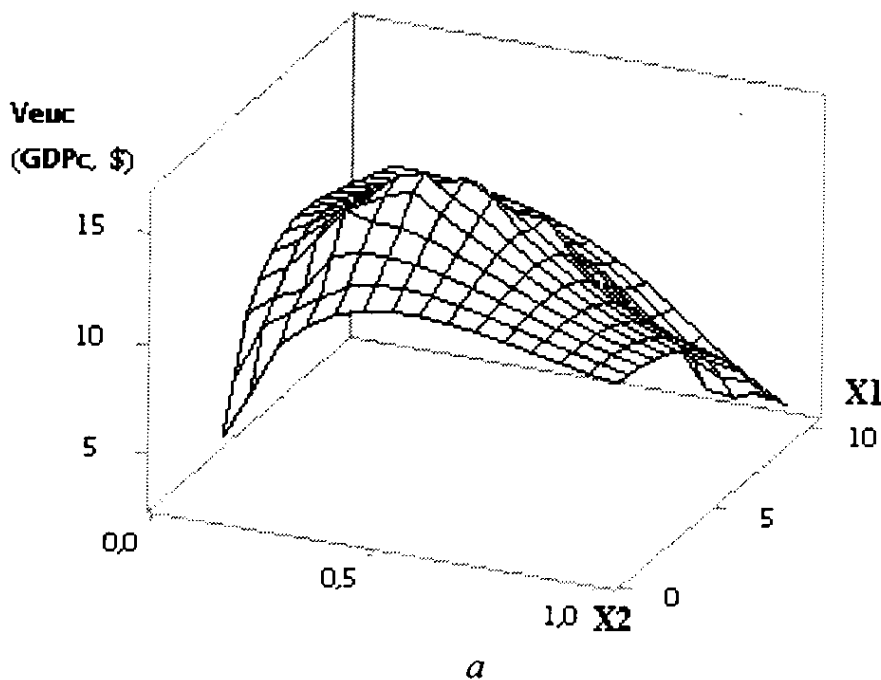
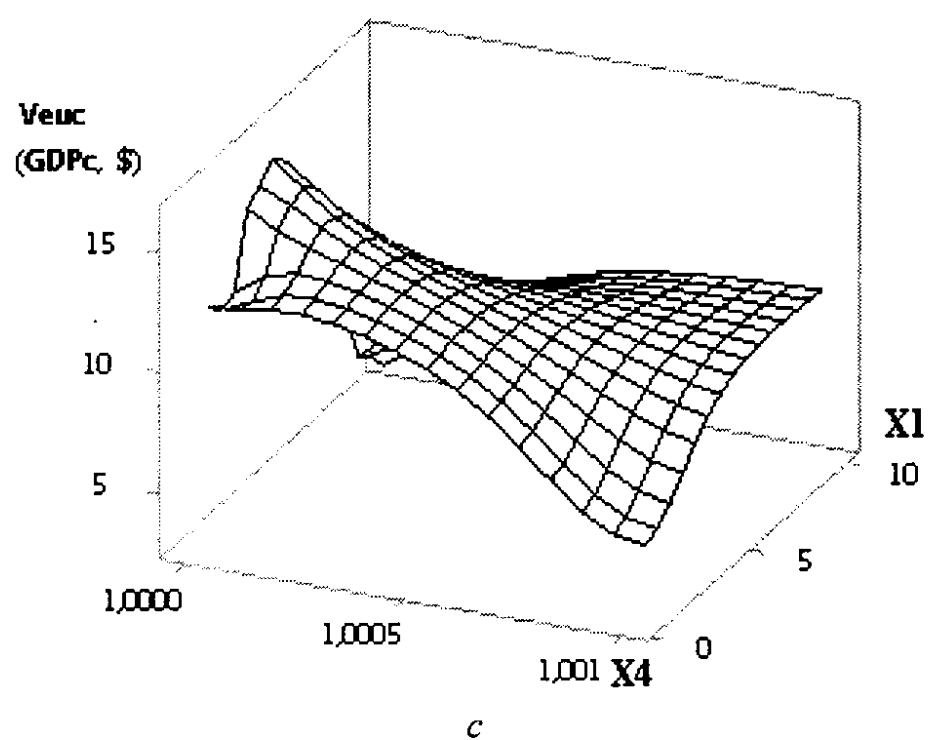
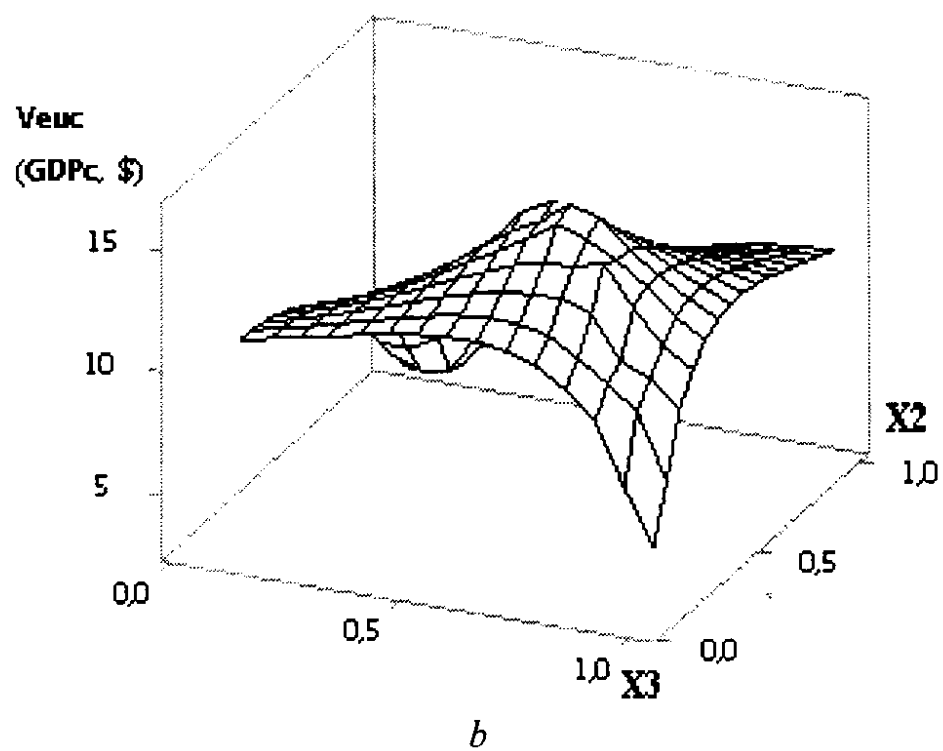


Fig. 5 Dependence $Veuc (GDPc) = f(X1, X2, X3, X4)$

when $X1 = 1...10$, $X2 = 0,99$, $X3 = 1...0,1$, $X4 = 1$





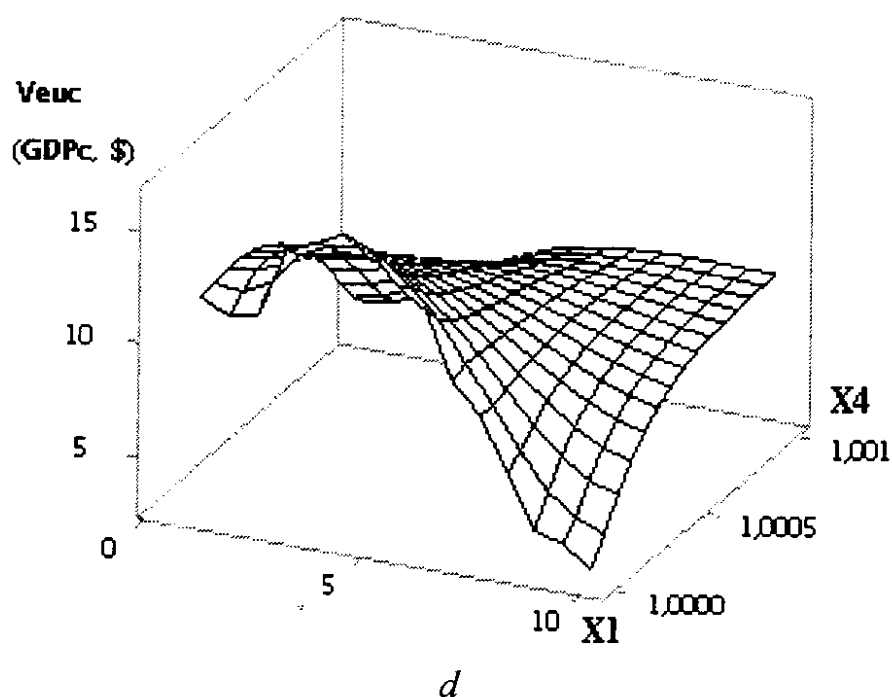


Fig. 6 3D-graphics:

a – $Veuc (GDPc) = f(X2, X1)$; *b* – $Veuc (GDPc) = f(X3, X2)$
c – $Veuc (GDPc) = f(X4, X1)$; *d* – $Veuc (GDPc) = f(X1, X4)$

when $X1 = 1...10$, $X2 = 0,1...0,99$, $X3 = 1...0,1$, $X4 = 1$

In the Fig. 7 there is a plotted curve V_{euc} which is almost identical to that in the Fig. 5, but in this case the maximum value of V_{euc} already equals 269, which is by 16.6 times more. Therefore, the following parameter values: $X1 = 4$, $X2 = 0.99$, $X3 = 0.7$ and $X4 = 1$ are more advantageous for use. The Fig. 8 shows two examples of 3D dependency $V_{euc} (GDPc) = f(X1, X2, X3, X4)$.

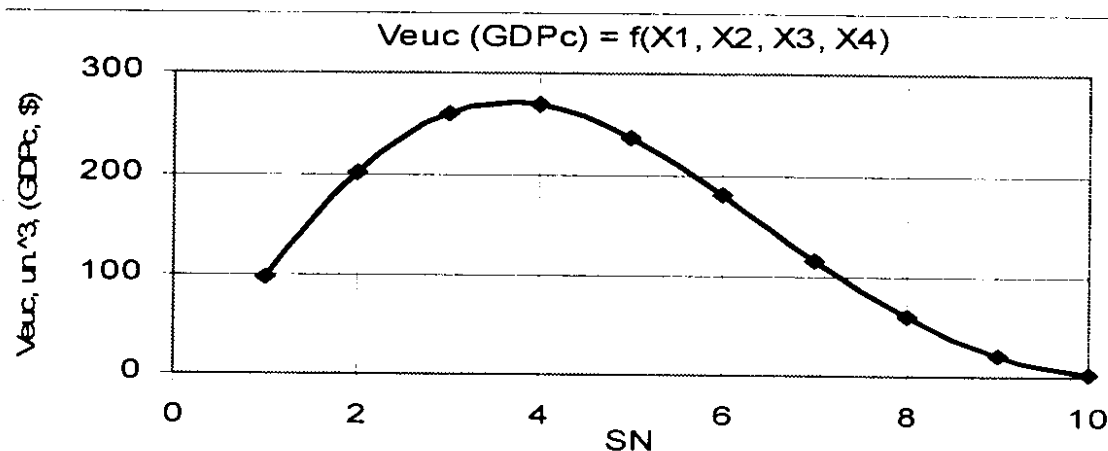


Fig. 7 Dependence $Veuc (GDPc) = f(X1, X2, X3, X4)$

when $X1 = 1 \dots 10$, $X2 = 0,99$, $X3 = 1 \dots 0,1$, $X4 = 1$

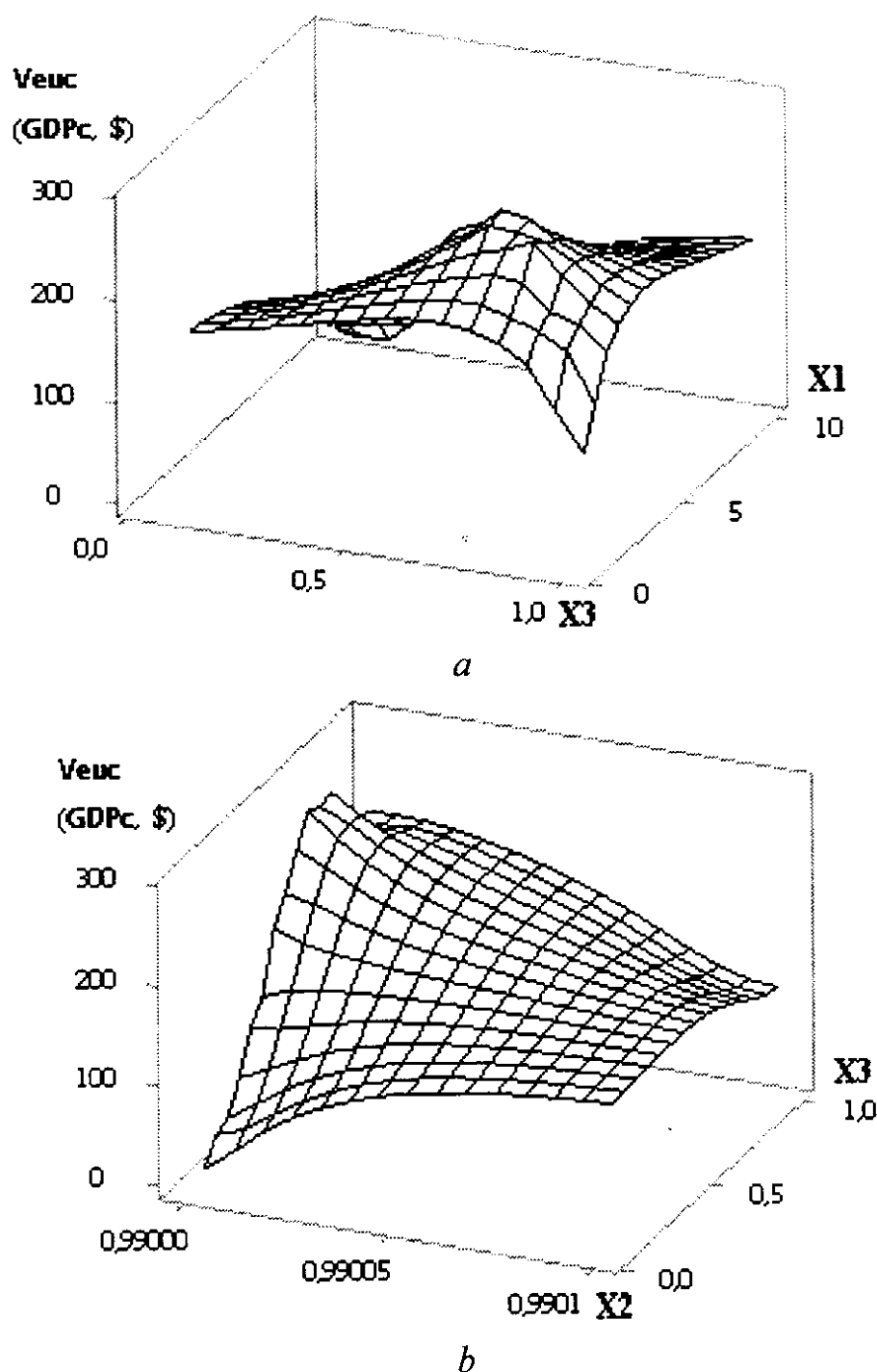


Fig. 8 3D-graphics:

$a - V_{euc} (GDPc) = f(X3, X1)$; $b - V_{euc} (GDPc) = f(X2, X3)$

when $X1 = 1 \dots 10$, $X2 = 0,99$, $X3 = 1 \dots 0,1$, $X4 = 1$

As can be seen from the Fig. 9, here the V_{euc} value is the largest of all the examples listed above, i.e. $V_{euc} = 8514.56$ at the point 7 with the

parameter values: $X_1 = 4$, $X_2 = 0.99$, $X_3 = 0.7$ and $X_4 = 1$. There is also one 3D image V_{euc} presented here (see Fig. 10).

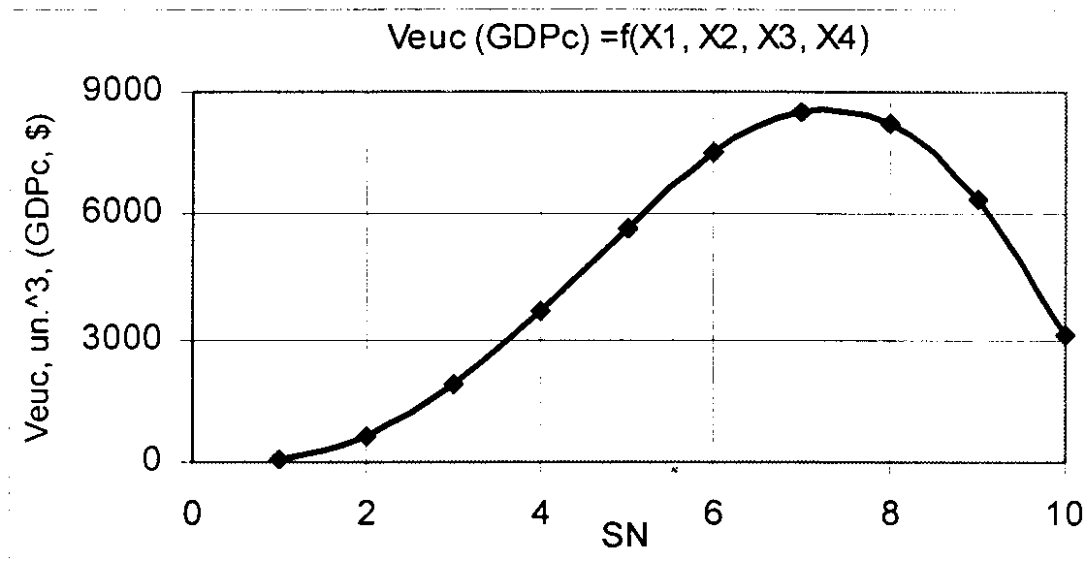


Fig. 9 Dependence $Veuc (GDPc) = f(X_1, X_2, X_3, X_4)$

when $X_1 = 1 \dots 0,1$, $X_2 = 0,99$, $X_3 = 1 \dots 10$, $X_4 = 1$

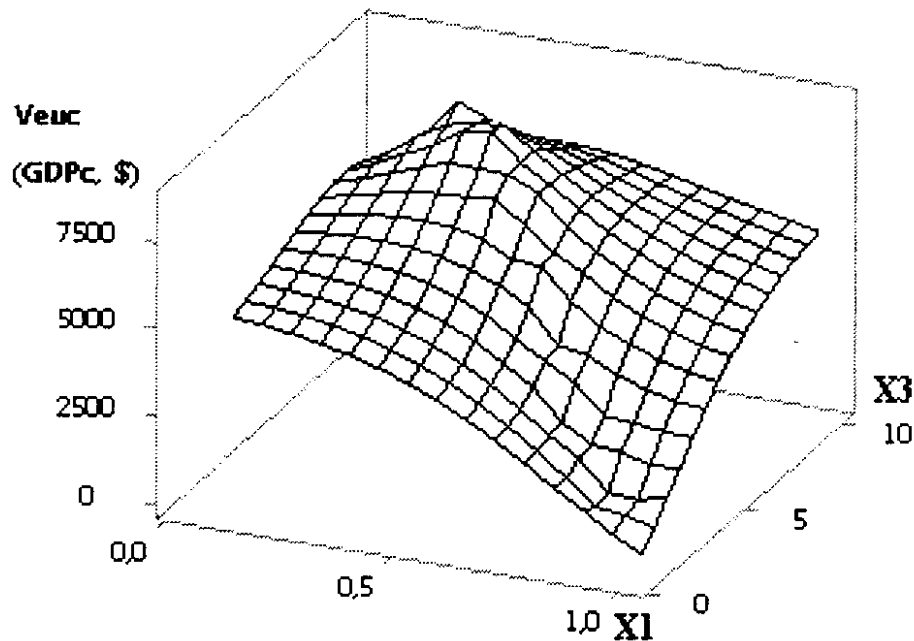


Fig. 10 3D-graphics: $Veuc (GDPc) = f(X_1, X_3)$

when $X_1 = 1 \dots 0,1$, $X_2 = 0,99$, $X_3 = 1 \dots 10$, $X_4 = 1$

In the following figure 11 it is seen that the plotted curve V_{euc} has a maximum value of $V_{euc} = 509.87$ and is symmetric regarding to its

maximum. The Figure 12 shows how the plotted curve V_{euc} changes in three-dimensional space.

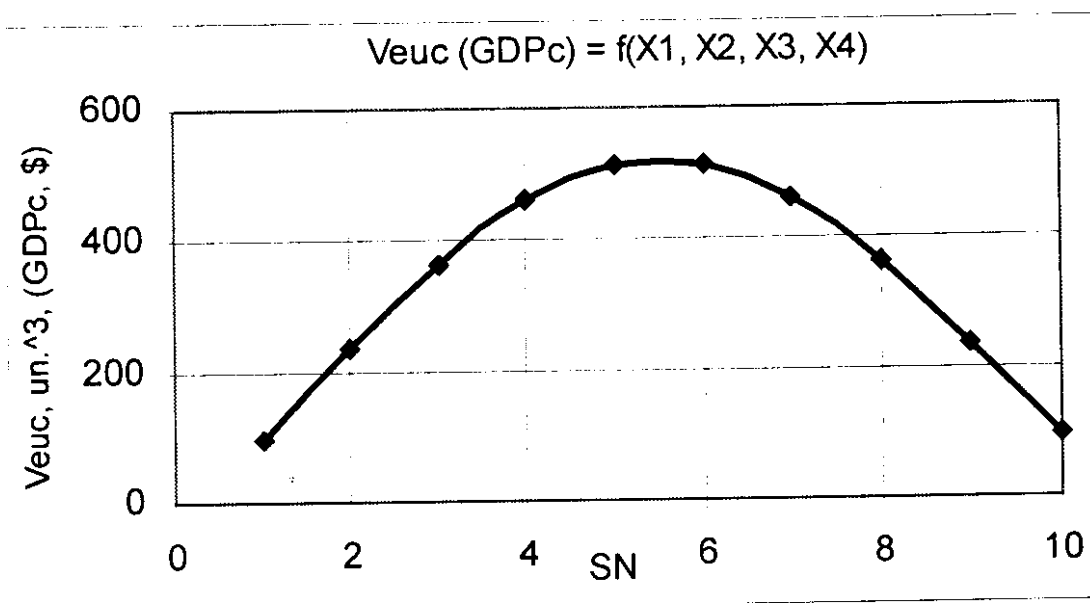


Fig. 11 Dependence $Veuc (GDPc) = f(X1, X2, X3, X4)$

when $X1 = 1...0,1$, $X2 = 0,99$, $X3 = X4 = 1...10$

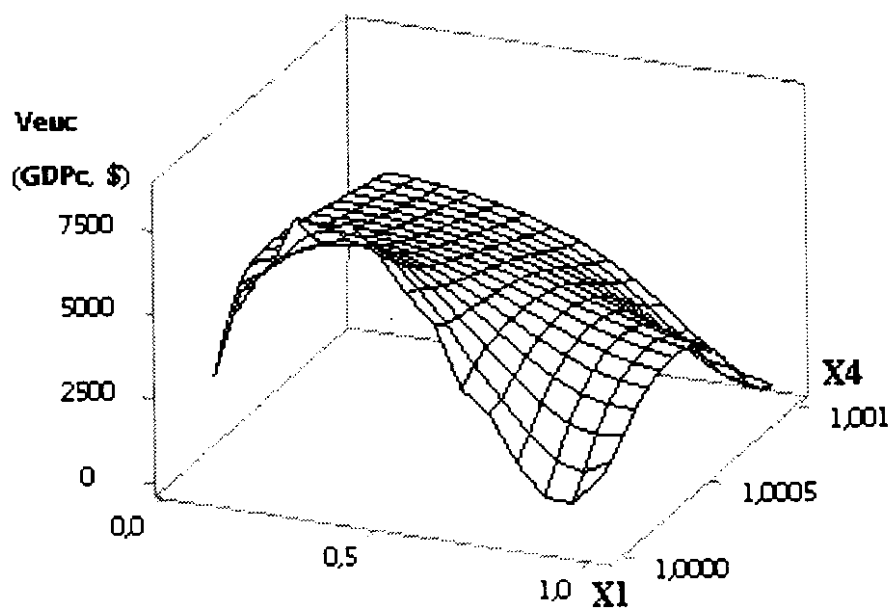


Fig. 12 3D-graphics: $Veuc (GDPc) = f(X1, X4)$

when $X1 = 1...0,1$, $X2 = 0,99$, $X3 = X4 = 1...10$

After the calculations have been made and their results were summarized in the overall table, which turned out to consist of 84 lines, despite the fact that 72 two-dimensional graphs were plotted. This is due to

the fact that a number of built graphs had maximums and minimums. In this summary table were introduced such relations as:

- $V_{euci} \dots V_{eucf}$, where V_{euci} – is the initial volume value of financial shell, unit³; V_{eucf} – is the final value of the volume of financial shell, unit³;
- V_{eucf} / V_{euci} – is the relation of financial shell final volume value to its initial value.

The relation of the financial shell final volume value V_{eucf} to the initial value of V_{euci} shows how many times the financial shell volume, i.e., GDP_c have increased (decreased) being affected by various external forces. Therefore, having such data, we can choose such parameters of X1, X2, X3 and X4, under which the financial shell volume will remain unchanged or even increase, being influenced by external forces. Thus, during the financial crisis, the selected parameter values will allow preserving the country's GDP_c at the same level, or even increasing it. After the table with 84 lines was built, it was transformed in the following way – only those values where $V_{eucf} / V_{euci} \geq 1$ were left. On the basis of this transformation the final table 1 presented below, has been obtained. The Table 1 shows only 15 scenarios from 43 that can be used upon the occurrence of the next financial crisis. As shown in Table 1 there are three scenarios under which the country's GDP_c will not change during the financial crisis that allow country's economy remain at pre-crisis level, and wait for the ending of the crisis. These are the scenarios where the relation $V_{eucf} / V_{euci} = 1$.

Table 1 - Statistics of theoretical relation V_{eucf} / V_{euci} where $V_{eucf} / V_{euci} \geq 1$

No. in sequence	X1, unit	X2, unit	X3, unit	X4, unit	$V_{euci} \dots V_{eucf}$, unit ³ ($GDP_{euci} \dots GDP_{eucf}$), \$	V_{eucf} / V_{euci} GDP_{eucf} / GDP_{euci}
1.	1	0,99	1...10	1...0,1	98,12...3,1E+06	31622,8
2.	1	0,99	1...10	1	98,12...98124,68	1000,0
3.	1...10	0,099...0,99	1	1	5,24...3102,97	592,5
4.	1...0,1	0,99	1...10	1	98,12...8514,56	86,8
5.	1	0,99...0,1	1...10	1	98,12...5237,53	53,38
6.	1	0,099...0,99	1	1...10	0,46...3,1	45,8
7.	1	0,99	1	1...0,1	98,12...3102,97	31,62
8.	1...10	0,99	1	1	98,12...3102,97	31,6
9.	1	0,099...0,99	1	1	5,24...98,12	18,73
10.	1	0,99...0,1	1	1...0,1	14,5...165,63	11,42

Table 1 continuation

11.	1...10	0,99...0,1	1	1	48,07...165,63	3,45
12.	1...0,1	0,099...0,99	1	1	1,52...3,1	2,04
13.	1	0,99	1	1	98,12...98,12	1,0
14.	1...10	0,99	1	1...10	98,12...98,12	1,0
15.	1...0,1	0,99	1	1...0,1	98,12...98,12	1,0

From all other scenarios it is reasonable to apply those where the maximum numbers of X1, X2, X3 and X4 parameters are the same, i.e. the units are equal. For example, scenario 9 has three parameters equal to one, and can increase the GDP_c by 18.73 times, while its X2 parameter increases by 10 times.

On the basis of the Table 1 (43 scenarios) analysis, Table 2 was constructed, showing which parameters were used, for how many times were they used and their percentage scenarios.

Table 2 - Statistics of parameters use, based on the table 1

Parameters	X1, unit	X1, unit	X1, unit	X2, unit	X2, unit	X2, unit
Values	1...10	1...0,1	1	0,099...0,99	0,99...0,1	0,99
Quantity	17	15	11	17	9	17
%	39,53	34,88	25,58	39,53	20,93	39,53
Parameters	X3, unit	X3, unit	X3, unit	X4, unit	X4, unit	X4, unit
Values	1...10	1...0,1	1	1...10	1...0,1	1
Quantity	25	5	13	9	17	17
%	58,14	11,63	30,23	20,93	39,53	39,53

REFERENCES

1. E. A. Pil. The boundaries of the existence of financial shells. Almanac of modern science and education.
2. E. A. Pil. Application of the theory of shells for describing the processes occurring in the economy. Almanac of modern science and education.
3. E. A. Pil. The effect of different variables on the financial shell of the country. Almanac of modern science and education.